Standard School Management Software

Feature List and Technical Specification

Reference model: a multi-campus, English-medium school following Pearson Edexcel curricula with an Islamic ethos, such as **Lighthouse International School** (Dhanmondi, Gulshan and Lalmatia campuses, 1,100+ students).

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This document lists **172 features across 21 modules**, each tagged by delivery phase, followed by the technical specification (architecture, stack, security, integrations, performance, hosting, migration and acceptance). It is intended to be read by the Principal, campus heads, the accounts and admission teams, and the school's IT adviser.

Contents

Section	Title
1	Purpose, scope and how to read this document
2	School context and assumptions
3	Solution overview: modules, users and phasing
4	Functional feature list (21 modules, 172 features)
5	Technical specification (5.1 – 5.14)
6	Delivery phasing and release plan
7	Assumptions, exclusions and questions for the school
A–C	Appendices: grading examples, report catalogue, notification catalogue

1. Purpose, Scope and How to Read This Document

A modern school management system replaces registers, spreadsheets, separate fee software and WhatsApp groups with one platform that every campus, teacher, parent and student uses. This document defines what a **standard, complete** system should do (functional features) and how it should be built and operated (technical specification), using Lighthouse International School as the reference model.

It can be used to: (a) agree scope internally; (b) request and compare vendor proposals on a like-for-like basis; (c) form the requirements baseline for development, testing and user acceptance.

Priority and phase tags

Tag	Meaning	Typical content
P1	Phase 1: needed on day one of go-live	Student records, admission, attendance, fees and online payment, exams and report cards, communication, parent app, core reports
P2	Phase 2: next release, within roughly 3 –6 months of Phase 1	Devices, transport, library, inventory, WhatsApp, Edexcel entries, payroll extensions, analytics
P3	Phase 3: later or optional	GPS tracking, procurement, online testing, advanced analytics, alumni, asset management

Feature IDs (for example **FEE-04**) are stable references for contracts, test cases and change requests.

2. School Context and Assumptions

The reference profile below is based on public information about Lighthouse International School and standard practice in comparable Dhaka schools. It must be validated with the school before scope is frozen.

Dimension	Assumption used in this specification
Organisation	A school of Akij Insaf with a single governing body and principal, operating as one organisation with several campuses.

Dimension	Assumption used in this specification
Campuses	Three campuses in Dhaka (Dhanmondi main campus, Gulshan, Lalmatia), with room for more. Each campus has its own head, office and accounts desk but shares policies, fee rules and reporting.
Students and staff	1,100+ students today; design headroom for 3,000+ students and 400 staff.
Curriculum	Pearson Edexcel pathway (primary and lower secondary, International GCSE, International A Level) alongside Bangla and Islamic Studies, with the school's own assessment calendar.
Ethos	Islamic values programme: prayer times in the routine, Quran and Islamic Studies progress tracking, Hijri calendar awareness, charitable activities.
Language	English as the medium of instruction; parents communicated with in Bangla and English.
Parents' devices	Smartphones (mostly Android) with mobile data; mobile banking (bKash, Nagad, Rocket) is the most common payment method.
Existing systems	Unknown; assumed to be spreadsheets, paper registers and possibly a basic fee tool. A migration plan is part of the scope (Section 5.12).
Website	The public website lighthouseschoolbd.com remains a separate, fast, SEO-optimised site that consumes live data from the school system through an API.

3. Solution Overview

3.1 Module map

The system comprises 21 modules on one shared platform. The table summarises the number of features by phase.

Code	Module	P1	P2	P3	Total
CORE	Platform, Administration & Security	11	2	0	13
ADM	Enquiry, Admission & Enrolment	8	2	0	10
SIS	Student Information System	7	2	1	10
ACA	Academic Structure, Timetable & Calendar	7	2	0	9
ATT	Attendance & Leave	6	2	0	8
LRN	Teaching, Learning & e-Classroom	4	4	1	9
EXM	Examination, Assessment & Reports	7	3	2	12
FEE	Fees, Billing & Payments	8	3	0	11
ACC	Accounts, Budget & Procurement (optional)	0	3	3	6
HRM	Staff, HR & Payroll	3	4	1	8
COM	Communication & Engagement	5	5	0	10
PRT	Parent, Student & Staff Portals and Mobile Apps	9	1	0	10
TRN	Transport Management	0	3	3	6
LIB	Library Management	0	4	2	6
INV	Inventory, Store, Uniform & Asset Management	0	3	3	6

Code	Module	P1	P2	P3	Total
HLT	Health, Welfare & Pastoral Care	2	5	0	7
CCA	Co-curricular, Houses, Events & Islamic Values Programme	1	5	2	8
SEC	Campus Security, Visitors & Safe Pick-up	0	3	2	5
DOC	Certificates, Letters & Document Generation	4	1	0	5
WEB	Public Website & CMS Integration	4	2	0	6
BI	Dashboards, MIS & Reports	3	3	1	7
	Total	89	62	21	172

3.2 Users

User group	Main needs	Primary channel
Principal, campus heads, governing body	Dashboards, approvals, campus comparison, results and finance overview	Web, mobile summary
Heads of section / department	Timetable, cover, assessment moderation, discipline, teacher performance	Web
Teachers	Attendance, homework, marks, lesson plans, messaging, leave	Teacher app, web
Admission and front office	Enquiries, applications, tests, certificates, visitors	Web
Accounts and cashiers	Invoices, collection, reconciliation, payroll posting, vouchers	Web
HR	Staff records, leave, payroll, appraisals	Web
Exam office	Exam schedule, seating, marks, report cards, Edexcel entries	Web
Librarian, store, transport, nurse, counsellor	Module-specific operations	Web, scanner app
Parents / guardians	Attendance, homework, fees, results, notices, messaging, leave	Parent app, SMS
Students (senior years)	Timetable, homework, resources, results	Student portal / app
IT administrator	Users, roles, settings, integrations, monitoring	Web admin

3.3 Key end-to-end workflows

Admission to enrolment: website enquiry → counsellor follow-up → online application and fee → test/interview → merit or waiting list → offer → acceptance and admission fee → student record, portal account and fee plan created automatically.

Monthly fee cycle: fee plan + concessions → bulk invoices → SMS/app notification with payment link → parent pays by bKash/Nagad/card → gateway webhook verifies and posts receipt → reminders for unpaid → reconciliation with settlement file → ledger posting and dashboards.

Examination cycle: exam calendar → seating and admit cards → mark entry by teachers → moderation by head of section → lock → grade calculation → report cards (bilingual PDF) → embargoed publication → parent notification → result analysis and improvement actions.

Daily attendance: teacher marks (or device captures) → absent/late parents alerted within minutes → leave applications reconcile automatically → thresholds trigger warnings and exam-eligibility flags.

4. Functional Feature List

Each module below lists its features with an ID, a description and the delivery phase (P1 / P2 / P3).

4.1 Platform, Administration & Security (CORE)

The shared foundation every other module sits on: one installation serving all campuses, with strict role-based access and a full audit trail.

ID	Feature	Description	Ph.
CORE-01	Multi-campus tenancy	One system, many campuses (Dhanmondi, Gulshan, Lalmatia and future ones). Every record carries a campus; users see only their campus unless granted group-wide access.	P1
CORE-02	Academic year and term management	Create academic years, terms/semesters and sessions; open, lock and roll over a year with one guided wizard (promotion, fee carry-forward, class re-creation).	P1
CORE-03	Role-based access control (RBAC)	Roles (Principal, Vice-Principal, Head of Section, Class Teacher, Subject Teacher, Accountant, Admission Officer, Librarian, Transport In-charge, Parent, Student, IT Admin). Permissions are per module, per action (view/create/edit/approve/export) and per campus.	P1
CORE-04	Approval workflows	Configurable multi-step approvals (leave, fee waiver, discount, purchase, certificate issue) with delegation, escalation timers and e-mail/SMS/push nudges.	P1
CORE-05	Audit trail	Who changed what, when and from where, for every sensitive entity (marks, fees, attendance, student data). Before/after values; tamper-evident; searchable; exportable.	P1
CORE-06	Bilingual interface (English / বাংলা)	Full UI, SMS templates, report cards and certificates available in English and Bangla; Bangla numerals and Unicode (Bijoy not required); Hijri date display optional.	P1
CORE-07	Master data and settings	Campuses, sections, shifts, subjects, grading schemes, fee heads, house names, holidays, SMS sender IDs, letterheads, signatures, numbering formats (admission no., roll no., invoice no.).	P1
CORE-08	Notification hub	One service sends in-app, push, SMS, e-mail and WhatsApp messages from templates with merge fields, retry, delivery status and per-guardian preferences.	P1
CORE-09	Global search	Single search box finds a student, guardian, staff member, invoice or receipt by name, ID, phone, NID/birth-certificate number or barcode.	P1
CORE-10	Data import / export	Excel/CSV templates with validation preview for students, staff, fees and marks; scheduled and ad-hoc exports to Excel/PDF/CSV honouring the user's permissions.	P1
CORE-11	Two-factor authentication and SSO	Optional TOTP / SMS-OTP 2FA for staff; Google/Microsoft sign-in for staff mailboxes; strong password policy, lockout and session management.	P2
CORE-12	Data privacy tooling	Consent records, access logs for minors' data, data-retention rules, anonymised test data, right-to-correct workflow for guardians.	P2
CORE-13	Backup, restore and archival	Automated daily encrypted backups with tested restore; old academic years archived read-only but still searchable.	P1

4.2 Enquiry, Admission & Enrolment (ADM)

From first enquiry to a seated, invoiced student: the full admission funnel for ~1,100+ students across three campuses and multiple year-groups.

ID	Feature	Description	Ph.
ADM-01	Enquiry capture	Enquiries from website form, phone, walk-in, social media and campaign QR codes land in one inbox with source tracking and campus/class interest.	P1
ADM-02	Lead pipeline and follow-up	Stages (Enquiry → Campus visit → Applied → Test/Interview → Offered → Admitted/Lost), tasks, reminders, call notes and conversion analytics per source and counsellor.	P1
ADM-03	Online application form	Mobile-friendly form with draft-save, document upload (birth certificate, previous report, photos, vaccination card), guardian details and application fee payment.	P1
ADM-04	Admission test and interview scheduling	Create test sessions per class, assign rooms and invigilators, generate admit cards, enter scores, schedule parent interviews, send reminders.	P1
ADM-05	Selection, merit list and waiting list	Rule-based shortlisting (age cut-off, test score, sibling, staff-ward priority), merit list, waiting-list promotion when a seat is released, seat availability dashboard by class and campus.	P1
ADM-06	Offer letter and acceptance	Templated bilingual offer letters with fee schedule, deadline and online accept/decline; automatic seat release on expiry.	P1
ADM-07	Admission fee and invoice	Admission/session fee invoice raised on acceptance, payable online or at cashier; enrolment confirmed only on payment (configurable).	P1
ADM-08	Enrolment to Student record	One click converts an admitted applicant into a student with admission number, class, section, roll number, portal account and fee plan; no re-typing.	P1
ADM-09	Transfer-in handling	Previous school, TC verification, equivalence mapping of curriculum, subject gap flags.	P2
ADM-10	Admission reports	Enquiries vs admissions, seat fill rate, source ROI, test performance, rejected/withdrawn reasons, year-on-year comparison.	P2

4.3 Student Information System (SIS)

The single, authoritative student record, linked to guardians, siblings, documents and history.

ID	Feature	Description	Ph.
SIS-01	Student profile	Personal data, photo, blood group, religion, nationality, birth-certificate/NID, addresses, previous school, languages, special needs, house, campus, class, section, roll.	P1
SIS-02	Guardian and family linking	Multiple guardians per student with relationship, occupation, income band, authorised-pickup flag and primary SMS contact; siblings linked across campuses for one family login and combined fee view.	P1
SIS-03	Document vault	Scanned/uploaded documents per student with type, expiry, verification status and access control; virus-scanned and stored outside the web root.	P1
SIS-04	Class, section and roll management	Bulk allocation, section balancing by gender/score, roll generation rules, mid-year transfer between sections or campuses with history retained.	P1

ID	Feature	Description	Ph.
SIS-05	Promotion and graduation	Rule-based year-end promotion (pass/fail/conditional), bulk promote with review screen, repeat/hold, IGCSE → IAL transition, graduation and alumni conversion.	P1
SIS-06	Withdrawal and transfer certificate	Withdrawal workflow with dues clearance check (fees, library, transport, store), TC issue and register entry.	P1
SIS-07	Special educational needs (SEN) register	Confidential SEN/learning-support records, individual education plans, access-arrangement flags for exams; restricted to named roles.	P2
SIS-08	Student ID cards	Design templates with photo, QR/barcode, RFID number; batch print for a class; lost-card reissue with fee.	P1
SIS-09	Student timeline	Chronological view of admission, attendance events, marks, discipline, health visits, awards and communications for a 360° view.	P2
SIS-10	Alumni directory	Graduate records with higher-study destinations, contact consent, alumni portal and event invitations.	P3

4.4 Academic Structure, Timetable & Calendar (ACA)

Programmes, classes, subjects, rooms and a clash-free timetable that survives teacher absence.

ID	Feature	Description	Ph.
ACA-01	Programme and curriculum setup	Define stages (Early Years, Primary, Lower Secondary, Pearson Edexcel International GCSE, International A Level) with their year-groups, subjects, assessment schemes and grading scales.	P1
ACA-02	Subject catalogue and combinations	Compulsory, optional and elective subjects; subject groups; IAL subject-choice combinations validated against rules; Bangla, Islamic Studies and language electives.	P1
ACA-03	Teacher–subject–class allocation	Assign teachers to class/section/subject with periods per week; load report highlighting overloaded or idle teachers.	P1
ACA-04	Timetable builder	Drag-and-drop weekly timetable with constraint checks (teacher/room/section clash, max periods, consecutive-period limits); optional auto-generation; shifts per campus.	P1
ACA-05	Substitution management	When a teacher is absent, the system lists free qualified teachers, assigns cover, notifies them and records cover periods for payroll/reporting.	P1
ACA-06	Rooms and resource booking	Classrooms, labs, hall, playground and AV equipment with booking calendar and conflict detection.	P2
ACA-07	Academic calendar	Terms, holidays (national, Islamic, school), exam windows, PTMs and events published to staff, parents and the website; iCal/Google Calendar feed.	P1
ACA-08	Daily routine and class-teacher view	Printable routines for classes, teachers and rooms; class-teacher dashboard showing today's periods, absentees and pending tasks.	P1
ACA-09	Scheme of work and syllabus tracking	Term-wise syllabus with coverage % reported by teachers and reviewed by heads of department.	P2

4.5 Attendance & Leave (ATT)

Fast, accurate attendance with immediate parent alerts and automation-friendly hardware support.

ID	Feature	Description	Ph.
ATT-01	Daily and period-wise attendance	Teacher marks present/absent/late/leave on web or phone in under a minute per class; period-wise attendance for senior years; edit window with audit.	P1
ATT-02	Biometric, RFID and QR capture	Import or real-time sync from ZKTeco/Hikvision-class devices, RFID gate readers and QR scanners; manual correction workflow.	P2
ATT-03	Instant absence / late alerts	SMS and push to the primary guardian when a child is absent or late (configurable cut-off); arrival/departure alerts from gate devices.	P1
ATT-04	Student leave applications	Parents apply from the portal with attachment (medical certificate); class teacher approves; calendar and attendance auto-updated.	P1
ATT-05	Attendance rules and thresholds	Minimum attendance % per term/exam eligibility, warning letters, automatic flag to head of section and parents.	P1
ATT-06	Staff attendance	Device/geo-fenced mobile punch, shift rules, late/early tracking, overtime, linked to leave and payroll.	P1
ATT-07	Staff leave management	Leave types and balances (casual, sick, earned, maternity, unpaid), request–approve flow, holiday-aware day counting, leave calendar for substitution.	P1
ATT-08	Attendance analytics	Class/section/campus trends, chronic absentees, day-of-week patterns, correlation with results.	P2

4.6 Teaching, Learning & e-Classroom (LRN)

Lesson plans, homework and digital resources in one place, so parents see what was taught and what is due.

ID	Feature	Description	Ph.
LRN-01	Lesson planning	Templates aligned to Edexcel specification topics and learning objectives; plans submitted to the head of department for approval; shared plan library.	P2
LRN-02	Homework and assignments	Teachers post work with due date and attachments; students submit files/photos online; teacher grades with comments; late-submission rules; parent visibility.	P1
LRN-03	Class diary / daily digest	Daily class diary visible in the parent app: topics covered, homework, reminders, uniform or kit notes.	P1
LRN-04	Learning resources	Upload notes, slides, videos and past papers by subject and class with access control and download tracking.	P1
LRN-05	Online classes integration	Schedule and launch Google Meet / Microsoft Teams / Zoom sessions from the timetable; attendance captured from the join log; recordings linked to the lesson.	P2
LRN-06	Question bank and quizzes	Tagged question bank (MCQ, short, structured), auto-marked quizzes with time limits, randomisation, result analysis per topic.	P2
LRN-07	Online test / mock exam delivery	Secure timed tests with browser lock-down option, auto-submission and teacher moderation of free-text answers.	P3
LRN-08	Gradebook for continuous assessment	Classwork, projects, practicals and quizzes weighted into term grades with moderation by heads of department.	P1
LRN-09	Learner progress and target tracking	Baseline, target and current grades per subject (e.g., Edexcel target vs working-at grade) with intervention flags.	P2

4.7 Examination, Assessment & Reports (EXM)

Supports both the school's own terminal exams and Pearson Edexcel external exam administration, with configurable grading and polished bilingual report cards.

ID	Feature	Description	Ph.
EXM-01	Exam scheduling	Exam types (class test, mid-term, final, mock, board practice), date-sheet per class/campus, clash checks for students with elective combinations, publish to portal and website.	P1
EXM-02	Admit cards and seating plans	Auto-generated seating plans with room/invigilator assignment, admit cards with photo and QR, student-wise seat lookup.	P1
EXM-03	Mark entry and moderation	Per-subject mark entry on web or Excel upload with component weights (written, practical, oral, CA), absent/exempt codes, double-entry verification, lock after approval and controlled re-open.	P1
EXM-04	Configurable grading schemes	Percentage → letter/GPA/descriptor mappings per programme: Edexcel 9–1 (International GCSE), A*–E (IAL), local GPA 5.0 scale, primary descriptors (Exceeding/Meeting/Working Towards).	P1
EXM-05	Report cards and progress reports	Bilingual templates per programme with marks, grades, teacher remarks, attendance, co-curricular and conduct; batch PDF generation; digital signature and QR verification.	P1
EXM-06	Transcripts and consolidated results	Multi-year transcript, subject-wise history, cumulative GPA, rank/percentile (optional per policy), result publication control with embargo date.	P1
EXM-07	Pearson Edexcel exam entries	Candidate register with candidate numbers, subject/unit entry codes, series (Jan/May-June/Oct-Nov), entry-fee calculation per candidate, export in the format required by the awarding body's entry system, access-arrangement flags.	P2
EXM-08	Predicted grades and UCAS-style references	Teacher-predicted grades with evidence, head-of-section approval, reference letter generation for university applications.	P2
EXM-09	External results import	Import published Edexcel results (CSV/Excel), compare with predictions and mocks, produce value-added and subject analysis for management.	P2
EXM-10	Result analysis	Pass %, grade distribution, subject/teacher/class comparisons, top performers, weak-topic heatmaps, year-on-year and campus-on-campus.	P1
EXM-11	Re-check and re-mark requests	Student/parent request, fee, status tracking and outcome recorded against the entry.	P3
EXM-12	Question-paper and answer-script custody	Secure upload of question papers with release time-lock, script receipt log and storage location tracking.	P3

4.8 Fees, Billing & Payments (FEE)

Accurate multi-campus fee management with real-time parent payments in Bangladeshi channels and clean reconciliation for the accounts team.

ID	Feature	Description	Ph.
FEE-01	Fee structure builder	Fee heads (tuition, admission, session, exam, lab, transport, activity, uniform, Edexcel entry) by campus, programme, class and month/term; one-time, recurring and conditional heads.	P1

ID	Feature	Description	Ph.
FEE-02	Concessions, scholarships and waivers	Sibling, staff-ward, merit and need-based discounts as % or amount with approval workflow, validity period and renewal rules; full audit.	P1
FEE-03	Invoice generation	Bulk monthly/term invoices with due date, previous balance, late fee and discounts; individual ad-hoc invoices; bilingual PDF and SMS/e-mail delivery.	P1
FEE-04	Online payment gateway integration	bKash, Nagad, Rocket, cards and internet banking via a Bangladesh aggregator (e.g., SSLCommerz, aamarPay, bKash PGW); payment-link in SMS; webhook-verified instant receipt.	P1
FEE-05	Counter collection	Cashier screen for cash/cheque/POS/bank-deposit with barcode/QR invoice lookup, partial payments, advance payment, day-end cash closing and handover.	P1
FEE-06	Receipts and statements	Auto-numbered receipts, family-wise ledger statement, due list by class/campus, fee clearance certificate for TC/exam admit card.	P1
FEE-07	Fines, reminders and dunning	Configurable late-fee rules; staged SMS/e-mail/WhatsApp reminders (T-3, due, +7, +15) and short-list for principal follow-up.	P1
FEE-08	Refunds and adjustments	Refund against withdrawn admission or overpayment with approval, mode and accounting entry.	P2
FEE-09	Bank reconciliation	Match gateway settlement files and bank statements to receipts; exception queue; unreconciled-item ageing.	P2
FEE-10	Fee analytics	Collection vs target, defaulters trend, concession cost, collection by channel/cashier, aging buckets and cash-flow forecast.	P1
FEE-11	Accounting integration	Post fee income, receivables and receipts to the general ledger (see ACC) with campus/cost-centre dimension; period lock respected.	P2

4.9 Accounts, Budget & Procurement (optional) (ACC)

Light accounting for school management, or integration to the school's existing accounting package.

ID	Feature	Description	Ph.
ACC-01	Chart of accounts and vouchers	Receipt, payment, journal and contra vouchers with campus and cost-centre dimensions; approval and period lock.	P2
ACC-02	Ledger, trial balance and statements	General ledger, party ledgers, trial balance, income & expenditure, balance sheet, cash book, bank book.	P2
ACC-03	Budget vs actual	Annual budgets per campus/department with variance and approval for overruns.	P3
ACC-04	Purchase requisition and PO	Departmental requisitions → approval → quotation comparison → PO → goods receipt → supplier bill and payment.	P3
ACC-05	Vendor management	Supplier master, performance notes, payment terms and TDS/VAT fields as per Bangladesh rules configured by finance.	P3
ACC-06	Third-party accounting connector	Export/API sync of vouchers to an existing accounting package; mapping table maintained by the finance team.	P2

4.10 Staff, HR & Payroll (HRM)

The whole employee lifecycle for teaching and non-teaching staff across campuses.

ID	Feature	Description	Ph.
HRM-01	Staff master	Personal, qualification, experience, certifications (e.g., teaching credentials), contract, bank, NID/TIN, emergency contact, documents with expiry alerts.	P1
HRM-02	Recruitment	Vacancy posting synced to the website careers page, applications, shortlisting, interview panels, scoring, offer letters.	P2
HRM-03	Onboarding and exit	Checklists (IT account, ID card, uniform, orientation), probation tracking, resignation, clearance and experience certificate.	P2
HRM-04	Payroll	Salary structures, allowances, deductions, overtime, cover-period pay, advance/loan, festival bonus, income-tax computation, payslips, bank advice file; locked monthly payroll with approval.	P1
HRM-05	Performance appraisal	Annual/periodic appraisal forms, lesson-observation records, student-feedback inputs, goals and development plans.	P2
HRM-06	Professional development (CPD)	Training catalogue, attendance, certificates and CPD-hours tracking per teacher.	P3
HRM-07	Staff self-service	Staff app/portal: payslips, leave, attendance, timetable, notices, profile updates with approval.	P1
HRM-08	HR analytics	Headcount, turnover, absenteeism, payroll cost by campus/department, teacher–student ratio.	P2

4.11 Communication & Engagement (COM)

Reliable, trackable communication between school, parents, students and staff, replacing scattered WhatsApp groups and paper circulars.

ID	Feature	Description	Ph.
COM-01	Notice board and circulars	Targeted notices by campus, class, section, house or individual with attachments; published to portal, app and (optionally) the public website; read receipts.	P1
COM-02	Bulk SMS with Bangla Unicode	Template-based SMS through Bangladeshi gateway APIs with masking sender ID, Unicode Bangla, segment/cost estimator, balance alerts and delivery reports.	P1
COM-03	E-mail and push notifications	HTML e-mail templates and mobile push for every system event; per-user preferences and quiet hours.	P1
COM-04	WhatsApp Business messaging	Approved-template messages (fee due, absence, result published) via WhatsApp Business API with opt-in tracking.	P2
COM-05	Parent–teacher messaging	Moderated one-to-one and class-level messaging inside the app, with office hours, retention and escalation to the head of section.	P2
COM-06	Parent–teacher meeting (PTM) scheduler	Teachers publish slots; parents book online; reminders; notes recorded against the student.	P2
COM-07	Surveys and feedback	Parent, student and staff surveys with anonymity option, analytics and action tracking.	P2
COM-08	Complaints and suggestions	Ticketed parent complaints with category, assignee, SLA, status and resolution notes; visible only to authorised roles.	P2
COM-09	Emergency broadcast	One-tap alert to all guardians of a campus/class via SMS and push (weather closure, security incident).	P1

ID	Feature	Description	Ph.
COM-10	Message templates library	Approved templates in English/Bangla with merge fields, version history and role-restricted editing.	P1

4.12 Parent, Student & Staff Portals and Mobile Apps (PRT)

Self-service on any phone, designed for busy parents; one family login for several children across campuses.

ID	Feature	Description	Ph.
PRT-01	Parent dashboard	Per child: today's attendance, homework, fees due, upcoming exams, notices, and a combined family view.	P1
PRT-02	Parent app (Android / iOS)	Native-quality app in English and Bangla with push notifications, biometric unlock and low-bandwidth mode.	P1
PRT-03	Fee payment in app	View invoices, pay by mobile banking/card, download receipts, see history and sibling-wise breakdown.	P1
PRT-04	Results and report cards	View marks and download report cards once published; trends and subject-wise analysis for parents.	P1
PRT-05	Student portal	Timetable, homework, resources, quizzes, results, library account, club sign-up and certificates for senior students.	P1
PRT-06	Teacher app	Mark attendance, post homework, enter marks, view timetable/cover duty, message parents, apply for leave.	P1
PRT-07	Profile updates with approval	Guardians request contact/address updates; office approves; history kept.	P1
PRT-08	Online forms and consents	Trip consents, photo/media consent, medical forms and policy acknowledgements with e-signature and reporting.	P2
PRT-09	Multi-child and multi-campus switch	Single login with a child switcher, regardless of campus.	P1
PRT-10	Accessibility and offline tolerance	WCAG 2.2 AA web, scalable fonts, high contrast, graceful behaviour on slow mobile data.	P1

4.13 Transport Management (TRN)

Safe, accountable school buses and vans with route planning, GPS visibility and fee integration.

ID	Feature	Description	Ph.
TRN-01	Routes, stops and vehicles	Route master, stops with pickup/drop times, vehicle register with capacity, fitness/insurance/tax-token expiry alerts, driver and helper assignment.	P2
TRN-02	Student transport assignment	Assign students to route/stop/vehicle, capacity checks, mid-year change with fee proration.	P2
TRN-03	Transport fee integration	Route/zone-based fees auto-added to invoices and cleared on withdrawal of service.	P2
TRN-04	GPS tracking and ETA	Live bus location and ETA to parents; geofence alerts for stop arrival, route deviation and speeding via supported GPS trackers.	P3
TRN-05	Boarding / alighting scans	RFID or QR scan on boarding and alighting with instant parent notification; exceptions list.	P3

ID	Feature	Description	Ph.
TRN-06	Fuel and maintenance log	Fuel consumption, service schedule, repair costs and downtime per vehicle.	P3

4.14 Library Management (LIB)

Catalogue, circulation and digital resources for libraries on each campus.

ID	Feature	Description	Ph.
LIB-01	Catalogue and accession	Books, journals and media with ISBN lookup, accession numbers, categories, shelf location, multiple copies and barcode labels.	P2
LIB-02	Issue, return and renewal	Barcode/RFID circulation with limits per role, due dates, renewal, reservations and e-mail/SMS reminders.	P2
LIB-03	Fines and clearance	Overdue and loss/damage fines posted to the fee ledger; library clearance for TC/exams.	P2
LIB-04	OPAC search for students	Online catalogue search with availability, reservation and reading history.	P2
LIB-05	Stock verification and reports	Periodic stock-take, missing/damaged books, circulation statistics, popular titles, reading levels.	P3
LIB-06	e-Library	Digital books and past-paper repository with access control by class and programme.	P3

4.15 Inventory, Store, Uniform & Asset Management (INV)

Stock and asset control across campuses, including uniform and book shop sales tied to student accounts.

ID	Feature	Description	Ph.
INV-01	Item master and stores	Items, units, categories and multiple stores (campus store, uniform shop, lab store) with min/max levels.	P2
INV-02	Goods receipt, issue and transfer	Stock-in from POs, issue to departments, inter-campus transfer with approval and batch/expiry where needed.	P2
INV-03	Uniform and book-shop POS	Size-wise stock, sale to student account or cash, sets/bundles by class, return/exchange and sales reports.	P2
INV-04	Fixed-asset register	Asset tagging with QR, location, custodian, depreciation, maintenance schedule, disposal and physical verification.	P3
INV-05	Lab consumables and equipment	Science/ICT lab inventory with breakage tracking and reorder alerts.	P3
INV-06	Maintenance requests	Staff raise facility/IT tickets with category, priority, assignee and SLA; photo evidence and completion sign-off.	P3

4.16 Health, Welfare & Pastoral Care (HLT)

Confidential health and pastoral records that keep children safe and keep parents informed.

ID	Feature	Description	Ph.
HLT-01	Health profile	Allergies, chronic conditions, medications, immunisations, emergency contacts and consent for treatment; visible to nurse and class teacher on a need-to-know basis.	P1
HLT-02	Sick-bay visits	Log symptoms, treatment, parent notification and sent-home decisions; alerts on repeated visits.	P2
HLT-03	Health check-ups	Annual height/weight/BMI/vision/dental records, camp scheduling and summary reports.	P2
HLT-04	Counselling and wellbeing	Confidential counsellor case notes, referral workflow, wellbeing surveys and flags for at-risk students.	P2
HLT-05	Behaviour and discipline	Merit/demerit points, incident reports, sanctions, parent communication and trend dashboards by student/class/staff.	P1
HLT-06	Safeguarding log	Restricted-access concerns log aligned to the school's child-protection policy, with escalation chain and audit.	P2
HLT-07	Anti-bullying cases	Case record, investigation steps, outcomes and follow-up reviews.	P2

4.17 Co-curricular, Houses, Events & Islamic Values Programme (CCA)

Beyond the classroom: clubs, sports, houses and the school's Islamic-values programme.

ID	Feature	Description	Ph.
CCA-01	Clubs and activities	Club catalogue, sign-up with capacity, attendance, achievements and activity fee.	P2
CCA-02	House system and points	Houses, captains, points from academics/sports/behaviour and live leaderboards.	P2
CCA-03	Sports day and competitions	Event categories, heats, results, certificates and medal tally.	P3
CCA-04	Field trips and events	Event planning, budget, consent forms, headcount, transport and post-event feedback.	P2
CCA-05	Awards and certificates	Template-based certificates (merit, attendance, participation) with QR verification, batch printing and portal download.	P1
CCA-06	Quran and Islamic studies progress	Configurable tracking of Quran recitation/memorisation milestones, Islamic Studies assessments and teacher notes for classes where the school runs these programmes.	P2
CCA-07	Prayer, Ramadan and Hijri calendar support	Prayer-time slots in the timetable, Ramadan routines, Hijri dates on calendar and holidays with moon-sighting adjustments.	P2
CCA-08	Charity and zakat/fitra collection (optional)	Voluntary donation drives with receipts kept separate from fees and reported to management.	P3

4.18 Campus Security, Visitors & Safe Pick-up (SEC)

Know who is on campus and who a child is released to.

ID	Feature	Description	Ph.
SEC-01	Visitor management	Pre-registration or walk-in with photo/ID capture, host notification, badge print and sign-out; watch-list.	P2

ID	Feature	Description	Ph.
SEC-02	Authorised pick-up	Guardians' authorised pick-up list with photo; QR/OTP at the gate for ad-hoc authorisations; pick-up log with timestamp.	P2
SEC-03	Gate entry / exit	RFID/QR gate logs for students and staff, late arrival slips and early-leave permits.	P2
SEC-04	CCTV and access-control linkage	Link events (e.g., gate scan) to the camera recording time for incident review; integration only, video stays in the NVR.	P3
SEC-05	Emergency drills	Drill schedule, roll-call by class using the app, evacuation reports.	P3

4.19 Certificates, Letters & Document Generation (DOC)

Every certificate and letter the office issues, generated from templates and verifiable online.

ID	Feature	Description	Ph.
DOC-01	Template designer	Bilingual templates with school letterhead, merge fields and digital signature/seal per campus.	P1
DOC-02	Standard certificates	Bonafide, character, transfer, testimonial, fee-paid, study, age and attendance certificates.	P1
DOC-03	Issue register and numbering	Auto serial numbers, issue log, reprint control and approval.	P1
DOC-04	QR verification page	Public page confirms that a certificate or report card is genuine (name, class, issue date only).	P2
DOC-05	Bulk letter generation	Mail-merge letters to parents (fee warnings, event invites) with PDF output and delivery tracking.	P1

4.20 Public Website & CMS Integration (WEB)

The public website (lighthouseschoolbd.com) stays fast and search-friendly while pulling live data from the school system.

ID	Feature	Description	Ph.
WEB-01	Notices, events and calendar feed	Published notices, events and holidays flow to the website automatically through an API; no double entry.	P1
WEB-02	Online admission and enquiry form	Website forms post directly into the ADM module with spam protection and campaign tracking.	P1
WEB-03	Result and certificate lookup	Public result lookup (with roll + DoB or OTP) and certificate verification pages.	P2
WEB-04	Careers page	Vacancies posted from HRM; applications flow to the recruitment pipeline.	P2
WEB-05	Gallery and news	Photo albums and news posts managed by the communications team, with image optimisation (WebP/AVIF) and consent flags.	P1
WEB-06	SEO and analytics	Structured data (School, Event, JobPosting), sitemap, per-page metadata, and campaign/goal tracking for admissions.	P1

4.21 Dashboards, MIS & Reports (BI)

Decision-ready views for the principal, campus heads and the governing body.

ID	Feature	Description	Ph.
BI-01	Role-based dashboards	Principal, campus head, accounts, admission, HR and teacher dashboards with live KPIs.	P1
BI-02	Standard report library	Student strength, attendance, fee collection, due list, result summary, staff roster, library and inventory reports; Excel/PDF export.	P1
BI-03	Campus comparison	Side-by-side KPIs across Dhanmondi, Gulshan and Lalmatia: enrolment, attendance, collection, results.	P1
BI-04	Ad-hoc report builder	Select fields, filters and grouping, save and share reports with permission checks.	P2
BI-05	Scheduled reports	E-mail daily/weekly/monthly reports (collection summary, absentee list) to chosen recipients.	P2
BI-06	Early-warning analytics	Flags for attendance decline, grade drop, fee delinquency and behaviour clusters to trigger intervention.	P3
BI-07	Regulatory and board returns	Pre-built exports for returns requested by education authorities, boards and the awarding body.	P2

5. Technical Specification

This section states the architecture, technology, security, integration and quality requirements for the school management system. The functional scope in Section 4 is stack-neutral; the stack below is the one EMSL recommends and can build and support, and every choice is open to review by the school's IT adviser.

5.1 Architecture principles

- **One platform, many campuses.** A single deployment serves all campuses; campus is a first-class dimension on every transactional record, not a separate installation.
- **Modular monolith first.** Modules (Section 4) are separate packages with clear internal APIs and their own database schema namespaces, deployed as one service. This keeps operations simple for a school-sized team and still allows a module to be extracted later.
- **API-first.** Web, mobile apps, the public website and third-party systems all use the same documented REST API (OpenAPI 3); no module reads another's tables directly.
- **Mobile-first, low-bandwidth tolerant.** Parents use phones on mobile data; pages and APIs are small, cached and resumable.
- **Secure by default, audited by design.** Least privilege, encryption in transit and at rest, and an append-only audit trail for sensitive changes.
- **Configuration over customisation.** Grading schemes, fee rules, workflows, templates and calendars are data that the school edits, not code that EMSL must change.
- **Bangla and English everywhere.** Unicode end to end, including SMS, PDFs, search and sorting.

5.2 Reference architecture

Layer	Components	Notes
Clients	Responsive web app (React PWA) for office and teachers; parent/student/teacher mobile apps (Android, iOS); public website (Next.js); kiosks and gate readers	All clients use the same REST API and OIDC sign-in
Edge	Cloudflare (DNS, CDN, WAF, DDoS protection, TLS), reverse proxy (Nginx) with rate limiting	Static assets cached at the edge; API never cached for private data
Application	ASP.NET Core modular monolith (Web API): Identity & Access, Core/Master data, Admission, SIS, Academics, Attendance, Learning, Exams, Fees, HR, Communication, Transport, Library, Inventory, Health, Reports	Stateless instances behind the proxy; horizontally scalable
Background work	Job scheduler and worker (Hangfire/Quartz.NET hosted services + queue): invoice runs, SMS dispatch, report-card PDF batches, device sync, backups, reminders	Idempotent jobs, retry with back-off, dead-letter queue and admin view
Data	PostgreSQL 16 (primary + streaming replica), Redis (cache, sessions, rate limits), S3-compatible object storage for documents, photos and PDFs	Row-level security by campus as defence in depth
Integration	Payment gateway adapters, SMS/e-mail/WhatsApp adapters, biometric/RFID device bridge, calendar and video-conference connectors, accounting connector	Each adapter isolated behind an interface; webhooks signed and replay-protected
Observability	Structured logs, metrics, traces and alerting (Prometheus, Grafana, Loki, Sentry-class error tracking)	Dashboards for payments, SMS delivery and job health

5.3 Technology stack (recommended)

Area	Choice	Why
Back-end	C# on .NET 10 LTS (or .NET 8 LTS), ASP.NET Core Web API (controllers + minimal APIs), ASP.NET Core Identity/Authorization policies, EF Core 9+ with Npgsql + Dapper for heavy reports, FluentValidation, EF Core migrations	Mature, long-term supported, strong transaction and security model for fees and marks; same language and tooling as EMSL's other products
Database	PostgreSQL 16 (UTF-8, Bangla collation), pg_trgm + full-text search for names	ACID, row-level security, JSONB for flexible forms, no licence cost
Cache / queue	Redis 7; RabbitMQ or a PostgreSQL transactional outbox (MassTransit/Wolverine) for reliable events	Fast sessions and rate limits; guaranteed delivery of notifications
Web front-end	React 19 + TypeScript (strict), Vite, TanStack Query/Router, React Hook Form + Zod, MUI or Radix-based component library with a custom school theme, PWA service worker, i18next (English/বাংলা)	Dense data-entry screens, keyboard navigation (Enter moves field to field), offline-tolerant
Mobile apps	React Native (Expo, TypeScript) for Android and iOS, FCM/APNs push, secure storage, biometric unlock; shares types, validation and API client with the React web app	One React skill set, shared code, small APK for low-end devices
Public website	Next.js (React; App Router, SSR/ISR) with WebP/AVIF images, JSON-LD, sitemap; reads public API endpoints only	Keeps the SEO gains of the redesigned website while showing live notices and events
Identity	OIDC / OAuth 2.1 (Keycloak or OpenIddict/Duende IdentityServer); short-lived JWT access tokens, rotating refresh tokens; optional Google/Microsoft staff SSO	Standards-based; single sign-on across web, apps and website admin
Documents & reports	QuestPDF or headless-Chromium rendering for bilingual PDFs (HarfBuzz Bangla shaping); ClosedXML for Excel; QR via QRCoder	Pixel-accurate report cards and certificates with Bangla shaping
Storage	S3-compatible object storage (cloud bucket or MinIO) with server-side encryption and signed, expiring URLs	Documents never served directly from web root
DevOps	Git, CI/CD (GitHub Actions or GitLab CI), Docker/Podman images, Infrastructure-as-code, blue/green or rolling deploys	Repeatable releases and fast rollback
Testing	xUnit + Testcontainers for .NET, Playwright end-to-end, k6 load tests, OWASP ZAP in CI, axe accessibility checks	Automated regression on fees, marks and permissions

> **Alternative stacks.** If the school or its IT team prefers a different back-end or database (for example SQL Server or Laravel), the functional requirements, security controls and non-functional targets in this document apply unchanged. The choice should weigh local support skills and hosting licence cost.

5.4 Multi-campus tenancy and core data model

The deployment is single-organisation (the Lighthouse group), multi-campus. Three scoping keys appear on records: **organisation_id** (future-proofing for sister schools), **campus_id** (Dhanmondi, Gulshan, Lalmatia, …) and **academic_year_id**. Users hold role assignments scoped to one campus, several campuses or the whole organisation.

Domain	Principal entities (illustrative)
Identity & access	User, Role, Permission, RoleAssignment(campus scope), Session, AuditEvent, ConsentRecord

Domain	Principal entities (illustrative)
Organisation	Organisation, Campus, AcademicYear, Term, Programme, YearGroup, Section, Shift, Room, Holiday
People	Student, Guardian, StudentGuardian, Sibling link, Staff, StaffContract, Document, HealthProfile, SenRecord
Admission	Enquiry, Application, TestSession, TestScore, Offer, WaitlistEntry, Enrolment
Academics	Subject, SubjectGroup, StudentSubject, TeacherAllocation, TimetableSlot, Substitution, LessonPlan, Homework, Submission
Attendance	StudentAttendance, PeriodAttendance, DeviceEvent, LeaveRequest, StaffAttendance, StaffLeave
Assessment	Exam, ExamSchedule, AssessmentComponent, Mark, GradingScheme, GradeBand, ReportCard, Transcript, ExternalEntry(Edexcel), PredictedGrade
Finance	FeeHead, FeePlan, Concession, Invoice, InvoiceLine, Payment, Receipt, Refund, GatewayTransaction, SettlementFile, LedgerEntry
Operations	Route, Stop, Vehicle, BookTitle, BookCopy, Circulation, Item, StockMovement, Asset, Ticket
Communication	Notice, MessageTemplate, OutboundMessage(channel, status), Survey, Complaint, PtmSlot

Design rules: every table has a surrogate key, created/updated audit columns and soft-delete where legal retention requires it; money is stored as integer paisa or fixed-precision decimal (never floating point); all timestamps are stored in UTC and displayed in Asia/Dhaka; marks and fee postings are append-only with reversing entries rather than overwrites.

5.5 Security and privacy requirements

Control	Requirement
Authentication	Passwords hashed with Argon2id or bcrypt; minimum length and breach-list check; lockout and CAPTCHA after repeated failures; OTP login option for parents; optional TOTP 2FA mandatory for Accounts, HR, IT Admin and Principal roles.
Authorisation	Server-side RBAC on every endpoint; campus scoping enforced in the service layer and by PostgreSQL row-level security; “break-glass” access to confidential records (safeguarding, SEN, counselling) is separately logged and reviewed.
Transport and data protection	TLS 1.2+ everywhere with HSTS; database and object-storage encryption at rest; secrets in a vault or environment-injected store, never in source control; sensitive fields (national IDs, health) additionally encrypted at column level.
Application security	OWASP ASVS Level 2 as the design target; protection against injection, XSS, CSRF, SSRF, IDOR and mass-assignment; strict Content-Security-Policy and security headers; file-upload type/size checks, antivirus scan and re-encoding of images.
Audit and monitoring	Append-only audit of logins, permission changes, marks, fees, attendance edits, exports and certificate issue; alerts for unusual export volume, off-hours admin activity and repeated failed logins.
Payments	Card and wallet data never touch the school system (hosted gateway pages or tokenised redirects); webhook signature verification and server-to-server transaction validation before marking an invoice paid; idempotent callbacks.
Children’s data	Data minimisation, parental consent for photos/media, restricted access to minors’ records, and retention/disposal schedules; compliance with Bangladesh data-protection and cyber-security legislation in force at go-live (to be confirmed with the school’s legal adviser).

Control	Requirement
Backup and recovery	Daily full and 15-minute incremental/WAL backups, encrypted, stored off-site in a second location; quarterly restore test with a written result.
Testing	Independent vulnerability assessment and penetration test before go-live and after major releases; findings rated and closed against agreed deadlines.

5.6 Role and permission matrix (default)

Defaults below ship with the system and can be changed by the IT Admin. V = view, E = enter/edit, A = approve, O = own record / own children only, – = no access.

Module	Principal / Campus Head	Head of Section	Teacher	Accounts	Admission	HR	Parent	Student
Admission	V A	V	–	V	V E A	–	O (own application)	–
Student records	V E A	V E	V (own classes)	V	V	–	O view, request edit	O view
Attendance	V A	V E	E (own classes)	–	–	–	O view / leave	O view
Marks and results	V A	V E A	E (own subjects)	–	–	–	O (when published)	O (when published)
Fees and payments	V A	–	–	V E A	V (admission fee)	–	O view / pay	O view
Staff, leave, payroll	V A	V (own team)	O (self-service)	V (payroll posting)	–	V E A	–	–
Health & safeguarding	V (restricted)	V (restricted)	V (alerts only)	–	–	–	O (own child)	–
Communication	V E A	V E	E (own classes)	E (fee notices)	E (admission)	E (staff)	V O	V O
Reports & dashboards	V all campuses	V section	V own	V finance	V admission	V HR	O	O
Settings & users	A	–	–	–	–	–	–	–

5.7 Integrations

Integration	Purpose	Method / standard	Phase
Payment aggregator(s)	Cards, bKash, Nagad, Rocket, internet banking	Hosted checkout + REST; signed webhooks; settlement-file import. Candidates: SSLCommerz, aamarPay, bKash Payment Gateway, bank gateways	1
SMS gateway	Alerts, OTP, fee reminders in Bangla Unicode	HTTPS API of a BTRC-licensed provider; masking sender ID; delivery receipts	1
E-mail	Receipts, reports, notices	SMTP/API (SES, SendGrid, Google Workspace) with SPF/DKIM/DMARC	1
WhatsApp Business	Template notifications and replies	WhatsApp Business Cloud API via BSP	2
Push notifications	Parent/teacher apps	Firebase Cloud Messaging / APNs	1

Integration	Purpose	Method / standard	Phase
Biometric / RFID / QR devices	Student and staff attendance, gate and bus scans	Vendor SDK or push-protocol (ZKTeco/Hikvision class), on-premise bridge service with store-and-forward	2
Video conferencing	Online classes and meetings	Google Meet / Microsoft Teams / Zoom APIs; join-log import	2
Google / Microsoft Workspace	Staff SSO, calendar feeds, Drive links	OIDC, Calendar API, iCal	2
Pearson Edexcel	Candidate entries and results	CSV/Excel import-export in the awarding body's published formats (no public API assumed); manual upload by the exams officer	2
Accounting package	Journal posting to the group's ledger	REST/file connector with mapping table, or built-in ACC module	2
GPS tracking	Bus location	Vendor REST/MQTT feed; map tiles via OpenStreetMap/Google Maps	3
Public website (Next.js)	Notices, events, admission, careers, result lookup	Public read-only API with API key + CDN caching; form posts with CAPTCHA	1
National ID / government systems	Optional verification	Only where an authorised API exists; not assumed	3

5.8 Mobile apps

- **Platforms:** Android 8+ and iOS 15+; APK size target under 30 MB; also installable PWA for devices that cannot use the store.
- **Apps:** one Parent & Student app (role-aware) and one Teacher & Staff app; optional Gate/Transport scanner app.
- **Behaviour:** English/Bangla, push notifications with deep links, biometric unlock, secure token storage, image compression before upload, retry queue for attendance and marks taken offline and synced on reconnection.
- **Store compliance:** privacy policy, data-safety declarations and child-safety requirements prepared by the school; accounts created by the school, not by self-registration of strangers.

5.9 Reporting and analytics

- Operational reports run against the primary or replica database with pagination and Excel/PDF export; heavy reports run as background jobs and notify the user when ready.
- Management dashboards read from pre-aggregated summary tables refreshed every 15 minutes (fees, attendance, enrolment, results) so they stay fast during peak hours.
- Every report honours the user's role and campus scope; exports are written to the audit log.
- A governed data dictionary documents each KPI definition (for example, "collection rate" and "attendance %") so every campus reads the same number.

5.10 Non-functional requirements (proposed targets)

Attribute	Target
Capacity	Sized for 3 campuses and 1,100+ students today, with headroom for 3,000 students and 400 staff, scalable to 10,000 students and 5 campuses by adding instances, not by redesign.
Concurrency	500 concurrent users in normal operation; burst of 300 online payments per hour on due-date days and 1,500 simultaneous parent logins at result publication.

Attribute	Target
Response time	Interactive pages p95 under 2 s on a typical Dhaka 4G connection; API p95 under 500 ms (excluding file generation); report-card batch of 1,000 PDFs within 10 minutes.
Availability	99.5% monthly (school-hours 07:00–20:00 Sat–Thu measured separately at 99.9% target), excluding announced maintenance windows outside school hours.
Recovery	RPO 15 minutes; RTO 4 hours; documented failover and restore runbooks.
Data integrity	Payment-to-invoice reconciliation exact to the paisa; no negative stock; duplicate-payment protection; marks locked after approval.
Usability	Common tasks (take attendance, collect a fee, enter marks) achievable in ≤ 3 minutes by a trained user; keyboard-first data entry; consistent navigation across modules.
Accessibility	WCAG 2.2 AA for parent/student portals and public pages; sufficient contrast, labels, focus order and screen-reader support.
Browser / device support	Latest two versions of Chrome, Edge, Firefox, Safari; Android and iOS as above; layouts from 360 px to 4K.
Localisation	English and Bangla UI; Bangla and English digits; Asia/Dhaka time zone; BDT currency with lakh/crore grouping (e.g., 6,60,000); Gregorian with optional Hijri display.
Maintainability	Automated tests covering fees, marks and permissions at ≥ 80% branch coverage; zero-downtime schema migrations where feasible; semantic versioning and release notes.
Scalability of notifications	Queue-based sending of ≥ 10,000 SMS/push per hour with back-pressure and per-provider rate limits.

5.11 Hosting, environments and deployment

Item	Specification
Hosting options	A) Cloud region close to Bangladesh (e.g., AWS Singapore/Mumbai, Google Cloud, Azure) — recommended for resilience; B) a Tier-III data centre in Dhaka with off-site backup; C) hybrid: application in cloud, device bridge on campus. Final choice depends on the school's data-residency and budget preferences.
Starting footprint	App tier 2 × (4 vCPU, 8 GB RAM); database 1 primary + 1 replica (4 vCPU, 16 GB RAM, 200 GB SSD); Redis 2 GB; object storage 500 GB growing ~150 GB/year; Linux (Ubuntu LTS); containers.
Environments	Development, Test/UAT (anonymised copy of real data), Staging (production-like), Production; separate credentials and gateways (sandbox in non-prod).
Release process	Pull-request review, automated tests and security scans, staged rollout; releases scheduled outside school hours; one-click rollback.
Monitoring	24×7 uptime checks, payment-failure and SMS-failure alerts, disk/CPU/DB alerts to named on-call contacts; monthly service report to the school.
Campus network	Reliable internet at each campus (primary + backup link); Wi-Fi for teacher tablets; local device bridge for biometric/RFID with store-and-forward; no inbound firewall holes.

5.12 Data migration and onboarding

- Inventory of current records (spreadsheets, paper registers, any existing software) and a mapping workbook agreed with the campus administrators.
- Import templates with validation (duplicates, missing guardians, invalid phone numbers, class/section mismatches) and a dry-run report before any load.

- Migrate current-year students, guardians, staff, fee plans and opening balances first; back-years' results and archived documents next, as a read-only history.
- Parallel run for at least one fee cycle: cashiers use the new system while the old registers remain as a cross-check; sign-off on reconciled totals.
- Parent onboarding: SMS invitation with one-time link, in-campus help desk during the first fee weeks, short Bangla video guides.

5.13 Quality assurance and acceptance

- Requirement traceability: every feature ID in Section 4 maps to test cases; UAT scripts executed by campus staff, not only by the vendor.
- Acceptance gates: (1) all Phase-1 "must" features pass UAT; (2) fee calculations and one month of invoices reconcile with the existing records; (3) load test meets Section 5.10; (4) penetration-test criticals/highs closed; (5) backup restore demonstrated; (6) staff trained and sign-off by the Principal and Head of Accounts.
- Defect severity and response times agreed in the support schedule; hyper-care for the first 30 days after each campus goes live.

5.14 Documentation, training and support

- Role-based user manuals and quick cards (English and Bangla), admin guide, API documentation (OpenAPI), data dictionary and runbooks.
- Training: administrators (2 days), teachers (2 × 2 hours per campus), accounts and admission (1 day each), parents' orientation sessions and recorded videos.
- Support: help desk during school hours (phone, WhatsApp, e-mail, ticket portal), named account manager, monthly review and an annual roadmap workshop.

Appendix A. Grading Scheme Examples (Configurable)

These are illustrative defaults that the exam office can edit; the exact bands and policies must be confirmed by the school and the relevant awarding body's documents.

Programme	Scale	Notes
Primary (Years 1–6)	Descriptors, e.g., Exceeding / Meeting / Working Towards, or A–E with comments	Skill-based; teacher remarks and effort/behaviour grid
Lower Secondary (Years 7–9)	Percentage + letter grade; optional progress vs target	Termly reports and end-of-year examinations
International GCSE (Pearson Edexcel)	9–1 grading, U for ungraded; internal marks mapped to predicted grade bands	Component-level marks for mocks; entries and results tracked per series
International A Level (AS / A2)	A*–E, U; unit marks and UMS conversion where applicable	AS and A2 units tracked separately; cash-in and resit entries
Local GPA (if used)	5.00 scale with grade points (A+ = 5.00 … F = 0.00)	Available for internal reporting or national-curriculum tracks

Appendix B. Standard Report Catalogue (Phase 1)

Area	Reports
Students	Strength by campus/class/section/gender/house; new admissions; withdrawals; guardian contact list; birthday list; missing-document report; sibling list
Attendance	Daily absentee list; monthly register; class/student attendance %; late arrivals; below-threshold students; staff attendance and leave summary
Academics	Timetable by class/teacher/room; teacher load; cover periods; homework completion; syllabus coverage
Examinations	Date sheet; seating plan; mark sheets; tabulation sheet; subject-wise result analysis; grade distribution; top performers; failed/at-risk list; report cards; transcripts
Fees	Daily collection (by cashier/channel/head); due list and ageing; defaulters; concession register; fee plan vs collected; refunds; gateway settlement reconciliation; receipt register
Admission	Enquiry funnel; source performance; test results; merit/waiting list; seat availability; conversion rate
HR	Staff roster; contract expiry; leave balances; payroll sheet and bank advice; payslips; overtime and cover pay
Management	Principal's dashboard; campus comparison; enrolment trend; collection vs budget; certificates issued; audit log summary

Appendix C. Notification Event Catalogue (Phase 1)

Event	Recipient	Channels
Student absent / late	Primary guardian	SMS, push
Leave request approved / rejected	Guardian, class teacher	Push, in-app
Homework assigned / due tomorrow	Student, guardian	Push, in-app
Notice or circular published	Targeted audience	Push, SMS (urgent), e-mail
Invoice generated / due in 3 days / overdue	Guardian	SMS, push, e-mail, WhatsApp (P2)
Payment received	Guardian, cashier	SMS, e-mail receipt, push
Exam schedule or admit card ready	Student, guardian	Push, SMS
Result / report card published	Student, guardian	Push, SMS
Admission offer / test date / document reminder	Applicant guardian	SMS, e-mail
Teacher cover assigned	Substitute teacher	Push, SMS
Leave request pending approval	Approver	Push, e-mail
Emergency broadcast	All guardians of campus/class	SMS, push (priority)

6. Delivery Phasing and Release Plan

A phased rollout reduces risk: Phase 1 delivers a complete day-to-day operating system for one pilot campus, then extends to the other campuses, before adding the P2 and P3 modules.

Release	Scope	Indicative duration	Outcome / exit criteria
0. Discovery	Workshops with each campus, fee and grading policies captured, data inventory, final scope and design sign-off	3–4 weeks	Signed requirement baseline, migration map, UI prototypes
1. Foundation (P1 core)	CORE, ACA, SIS, ADM, ATT, COM, PRT (parent app), BI essentials	12–14 weeks	Pilot campus live for attendance, notices, student records, admissions
2. Money and results (P1 finance & exams)	FEE with online payments, EXM and report cards, DOC, WEB integration, HRM essentials (staff, leave)	8–10 weeks	Fee cycle run end-to-end in parallel; report cards produced for a term; all campuses live
3. Operations (P2)	Devices (attendance, gate), TRN, LIB, INV, HLT, CCA, SEC, WhatsApp, Edexcel entries, payroll extension, ACC connector	12–16 weeks (can be sequenced per module)	Operational modules live as campuses are ready
4. Optimisation (P3)	GPS, procurement, online testing, analytics, alumni, assets	Roadmap-driven	Selected on value after 6 months of use

> Durations are indicative for planning and depend on how quickly data, policies and approvals are available. A fixed plan is produced at the end of Discovery.

7. Assumptions, Exclusions and Questions for the School

Assumptions

- The school provides a decision-maker per campus and a single product owner with authority to approve scope and UAT.
- Each campus has dependable internet; the school procures its own SMS masking sender ID, payment-gateway merchant accounts, WhatsApp Business account and developer accounts for app stores.
- Pearson Edexcel exam administration continues to use the awarding body's own systems; this software prepares and tracks the data but does not replace them.
- Fee policies, grading policies and discount rules are provided in writing before configuration.

Exclusions (unless added by change request)

- Hostel/boarding management, canteen/meal billing, learning-content authoring (courseware) and a full learning-management platform replacing the school's existing teaching tools.
- Hardware (biometric devices, RFID readers, GPS trackers, tablets, servers) and campus networking; the software integrates with supported devices.
- Statutory legal opinions, accounting-policy advice and third-party licence fees (SMS, payment, WhatsApp, cloud hosting).

Questions to confirm with the school

#	Question	Why it matters
1	Which year-groups and programmes run at each campus (e.g., IAL at Dhanmondi only)?	Drives structure, fees, timetable and exam modules
2	Current student, staff and campus numbers, and expected growth over 3 years?	Sizing and licensing

#	Question	Why it matters
3	What is used today (software, spreadsheets, paper) for admission, fees, results and attendance?	Migration effort and integration needs
4	Which payment channels must be offered, and which bank/aggregator relationships already exist?	Gateway contracts and settlement flow
5	Is any biometric/RFID device installed or planned? Brand and model?	Device integration scope
6	Which grading rules and report-card formats must be reproduced exactly?	Report-card templates and calculation tests
7	Is transport run by the school, and is GPS tracking wanted?	TRN module phase
8	Hosting preference (cloud vs Dhaka data centre) and any data-residency rule from the governing body?	Hosting design and cost
9	Who owns the Islamic-values programme tracking and what must be measured?	CCA configuration
10	Which existing accounting software must receive fee postings?	ACC connector scope